Data Structures & Algorithms (CST-003) - Unit 3

Unit 3 - Linked Lists - notes in easy English, matched to the UTU syllabus.

Study Materials by Anas - study-materials-by-anas.vercel.app

What this unit covers: linked lists - singly, doubly and circular - and how insertion and deletion work without shifting elements.

Key ideas: Each **node** holds data plus a pointer to the next node; the list is reached from a **head** pointer. Advantages over arrays: no fixed size, O(1) insert/delete once you are at the spot. Disadvantages: no random access (must walk from head, O(n)), extra memory per node. Master the pointer steps for: **insert** at beginning / middle / end, and **delete** a node by value - draw the pointers before writing code; the order of pointer updates is what examiners check. A **doubly linked list** adds a prev pointer (walk both directions, more memory); a **circular list** links the last node back to the head.

Exam pointers: "write a function to insert/delete a node" is the standard coding question - practise the singly-linked versions until the pointer order is automatic. Array vs linked list comparison is a favourite theory 5-marker.